

The flow of blood through the human body is typically laminar, but in certain conditions, as when blood flows through the aorta during a period of heightened physical activity, the flow can become turbulent. Blood flow can also become turbulent in abnormal cases in which the passage of blood is obstructed by constrictions within the blood vessel. The energy generated by blood turbulence creates sound vibrations that can be heard with a stethoscope.

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Turbulent kinetic energy (TKE) is the kinetic energy per unit mass associated with turbulence within flowing fluids and is defined as:

$$TKE = \frac{1}{2} \sigma_v^2$$

Equation 1

where σ_v is the standard deviation of the fluctuating fluid velocities at a point within the turbulent flow.

Reynolds number Re is the quantity that ultimately measures whether a flow is turbulent or laminar. Re is defined in terms of the fluid density ρ , mean velocity of the flow $\bar{v}$, conduit diameter d , and dynamic viscosity η of the fluid, as given by:

$$Re = \frac{\rho \bar{v} d}{\eta}$$

Equation 2

In the context of blood flow, d is the diameter of the blood vessel, and blood has a fluid density of about $1,060 \text{ kg} \cdot \text{m}^{-3}$ with a dynamic viscosity of about $0.003 \text{ kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$.

Which of the following units can be used to express Reynolds number?

- ☐ A. $\frac{\text{g} \cdot \text{cm}}{\text{s}}$ [4%]
- ☐ B. $\text{kg}^2 \cdot \text{m}^2 \cdot \text{s}^2$ [9%]
- ☐ C. $\frac{\text{kg} \cdot \text{m}}{\text{s}}$ [24%]
- ☒ D. $\frac{\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}}{\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}}$ [61%]

Explanation:

Physical quantities always have dimensions (eg, length, time, mass, temperature) and are expressed in terms of **units**. In contrast, **pure numbers** are dimensionless and have no units. **Dimensional analysis** refers to the process by which the units of one variable featured in an equation can be determined algebraically using the units of the other variables within the equation.

Specifically, the units of Reynolds number Re are determined from the units of the fluid density ρ , mean velocity of the flow $\bar{v}$, conduit diameter d , and dynamic viscosity η of the fluid

$$Re = \frac{\rho \bar{v} d}{\eta}$$

In addition to the standard SI units for velocity (m/s) and length (m), the units of ρ and η are provided in the passage:

$$\left[\rho \right] = \frac{\text{kg}}{\text{m}^3}$$

$$\left[\eta \right] = \frac{\text{kg}}{\text{m} \cdot \text{s}}$$

where the bracketed terms denote units of the given variable.

Inserting each of the units above into the equation for Re yields:

$$\left[Re \right] = \frac{\left(\frac{\text{kg}}{\text{m}^3} \right) \cdot \left(\frac{\text{m}}{\text{s}} \right) \cdot (\text{m})}{\frac{\text{kg}}{\text{m} \cdot \text{s}}}$$

This simplifies to:

$$\left[Re \right] = \frac{\frac{\text{kg}}{\text{m} \cdot \text{s}}}{\frac{\text{kg}}{\text{m} \cdot \text{s}}} = \frac{\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}}{\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}}$$

As seen by the dimensional analysis, Re is a ratio of two quantities with the same dimension (**Choice D**). Consequently, the units in the numerator and denominator of the fraction defining Reynolds number completely cancel out, and Re is a **pure number** and **dimensionless**.

Educational objective:

Variables either have a dimension and are expressed in terms of a system of units, or they are dimensionless, pure numbers. Dimensional analysis is used to determine the possible units of a variable in an equation.

Time spent: 0 sec

Subject:
Physics

QID: 402657

System:
Fluids

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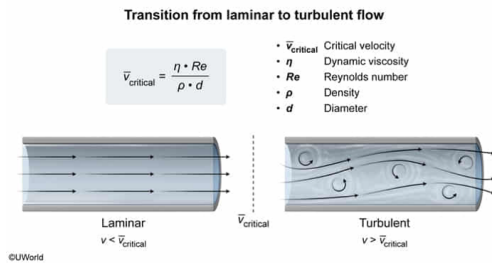
Blood flow in the aorta becomes turbulent at a Reynolds number of approximately 6,000. If the diameter of the aorta is approximately 2 cm, what is the average velocity of blood at this Reynolds number?

- ☐ A. 8.50 cm/s [16%]
- ☒ B. 85.0 cm/s [62%]
- ☐ C. 101 cm/s [15%]
- ☐ D. 170 cm/s [4%]

Correct

63% answered correctly

Explanation:



A flowing fluid can be conceptualized as a collection of packets traveling within the same conduit (ie, a pipe or container through which fluid flows). The flow of these packets can be described as either **laminar** or **turbulent**. In the case of laminar flow, the packets move together in a layered, uniform manner, and the flow is smooth. In cases of turbulent flow, the motion of the packets becomes highly nonuniform, and the resulting flow is more chaotic.

The transition from laminar to turbulent flow in a fluid is indicated by the **Reynolds number Re** of a flow. Large values of Reynolds number (ie, $Re > 2,000$) are associated with the onset of turbulence because, like the onset of turbulence, Re depends on both the characteristics of the fluid and the fluid flow.

In blood flows, the value of Reynolds number at which the transition between laminar and turbulent flow takes place is measured to be about 6,000. A rearrangement of Equation 2 may be used to express the velocity (v_{critical}) associated with the transition:

$$v_{\text{critical}} = \frac{\eta \cdot Re}{\rho \cdot d}$$

where η is the **dynamic viscosity** of blood, ρ is the **density** of blood, and d is the **diameter** of the vessel through which blood flows.

In this question, the diameter of the conduit (a blood vessel) is 2 cm (0.02 m), and the passage states that the density and viscosity of blood are 1,060 kg/m³ and 0.003 kg/m·s, respectively. Substituting these values into the equation above yields the value of v_{critical} :

$$v_{\text{critical}} = \frac{\left(0.003 \frac{\text{kg}}{\text{m} \cdot \text{s}}\right) \cdot 6000}{\left(1,060 \frac{\text{kg}}{\text{m}^3}\right) \cdot 0.02 \text{ m}}$$

The value of v_{critical} can be approximated as:

$$v_{\text{critical}} \approx \frac{(3 \times 10^{-3}) \cdot (6 \times 10^3)}{(1 \times 10^3) \cdot (2 \times 10^{-2})} \frac{\text{m}}{\text{s}} = \frac{18}{20} \frac{\text{m}}{\text{s}} = 0.90 \frac{\text{m}}{\text{s}} = 90.0 \frac{\text{cm}}{\text{s}}$$

which is slightly larger than the exact value:

$$v_{\text{critical}} = 0.850 \frac{\text{m}}{\text{s}} = 85.0 \frac{\text{cm}}{\text{s}}$$

Therefore, in this case the blood flow will transition from a laminar to a turbulent flow at a velocity of 85.0 cm/s (**Choice B**).

Educational objective:

Reynolds number indicates the transition from laminar to turbulent flows. The critical velocity at which the onset of turbulence occurs can be calculated using the Reynolds number, the conduit diameter, and

the density, as well as the dynamic viscosity, of the fluid.

Time spent:0 sec

Subject:
Physics

QID:402658

System:
Fluids

The flow of blood through the human body is typically laminar, but in certain conditions, as when blood flows through the aorta during a period of heightened physical activity, the flow can become turbulent. Blood flow can also become turbulent in abnormal cases in which the passage of blood is obstructed by constrictions within the blood vessel. The energy generated by blood turbulence creates sound vibrations that can be heard with a stethoscope.

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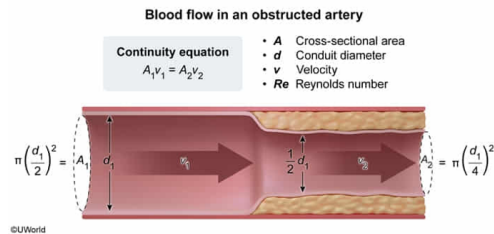
Equation 2

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The diameter of a segment of an artery is reduced by a factor of two due to an obstruction. Assume that the flow is incompressible and laminar, and therefore follows the continuity equation. Compared to an

unobstructed segment of the artery, the velocity of blood in the obstructed segment of the artery is:

- ☐ A. 1/4 as large. [8%]
- ☐ B. 1/2 as large. [13%]
- ☐ C. 2 times as large. [46%]
- ☒ D. 4 times as large. [31%]

Explanation:

The flow of a fluid refers to the motion of many individual particles that make up the fluid. If the flow lines used to conceptualize the flow of the particles remain parallel to each other and do not change over time (ie, each of the fluid layers move past each other with little friction), the flow is referred to as **laminar**.

For **incompressible** laminar flows traveling at velocity v between two regions of a conduit (eg, pipe) with **cross-sectional area** A , the quantity of mass entering one portion of a pipe over a period of time must equal the quantity of mass exiting that region simultaneously during that period. This relationship is stated by the continuity equation:

$$A_1 v_1 = A_2 v_2$$

For a circular conduit (eg, pipe, blood vessel), A can be expressed in terms of the diameter d :

$$A = \pi \cdot \left(\frac{d}{2}\right)^2 = \frac{\pi}{4} \cdot d^2$$

Substituting the expression for A into the continuity equation and then simplifying yields the relationship between v_1 and v_2 in terms of the respective diameters:

$$\frac{\pi}{4} \cdot (d_1)^2 \cdot v_1 = \frac{\pi}{4} \cdot (d_2)^2 \cdot v_2$$

$$v_2 = \left(\frac{d_1}{d_2}\right)^2 \cdot v_1$$

In this question, the diameter of the blood vessel is halved as it flows from one segment to the next ($d_2 = 0.5 \cdot d_1$). Using these proportions, the relationship between v_1 and v_2 simplifies to:

$$v_2 = \left(\frac{d_1}{0.5 \cdot d_1}\right)^2 \cdot v_1 = 4 v_1$$

Therefore, the continuity equation implies v is **inversely proportional** to the square of conduit diameter and *increases* by a *factor of four* as conduit diameter is halved.

(Choices A and B) v in a laminar flow is inversely proportional to the square of flow diameter. Therefore, velocity must *increase* if the diameter of the flow is reduced.

(Choice C) v is inversely proportional to the square of diameter, not the diameter. Consequently, v must increase by a factor of four.

Educational objective:

The continuity equation expresses the conservation of mass observed as fluids flow through a conduit. The continuity equation indicates that velocity is inversely proportional to the square of diameter.

The flow of blood through the human body is typically laminar, but in certain conditions, as when blood flows through the aorta during a period of heightened physical activity, the flow can become turbulent. Blood flow can also become turbulent in abnormal cases in which the passage of blood is obstructed by constrictions within the blood vessel. The energy generated by blood turbulence creates sound vibrations that can be heard with a stethoscope.

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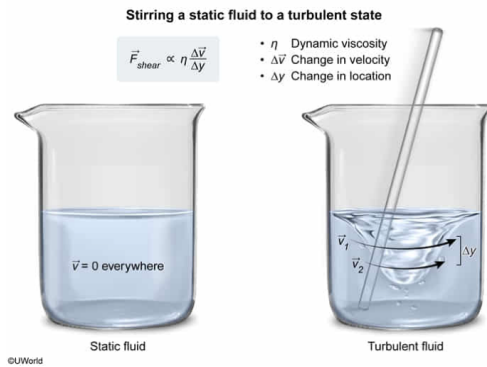
A fluid is stirred until it becomes turbulent. After the stirring stops, the fluid gradually returns to its original static state. The turbulent flow does not continue indefinitely because:

- ☐ A. the fluid is incompressible. [3%]
- ☐ B. the kinetic energy in the flow is transferred to gravitational potential energy. [20%]
- ☐ C. the energy put into the flow by stirring is transferred by convection. [12%]
- ✓ ☒ D. the energy put into the flow by stirring is dissipated by the fluid's viscosity. [63%]

Correct

63% answered correctly

Explanation:



Viscosity is an intrinsic property of a fluid that characterizes the amount of friction resisting motion inside the fluid itself. The friction due to viscosity arises from interactions between the molecules in a liquid or gas. The viscous friction force can be conceptualized as a shearing force between fluid layers that are flowing past each other with different velocities ($\vec{v}_1$ and $\vec{v}_2$).

The shearing force $\vec{F}_{\text{shear}}$ is proportional to gradient of the velocity $\Delta \vec{v}$ between each layer, which is a value that characterizes how the velocity changes within the fluid between different locations separated by some distance Δy :

$$\vec{F}_{\text{shear}} \propto \eta \frac{\Delta \vec{v}}{\Delta y} = \eta \frac{\vec{v}_2 - \vec{v}_1}{\Delta y}$$

where the coefficient η is the dynamic viscosity. The more viscous the fluid, the larger the coefficient η and the larger the shear force for given velocity gradients.

As a static fluid is stirred up, energy is transferred into the kinetic energy associated with the motion of the fluid. At first, the flow is laminar but it eventually becomes turbulent as the velocity of the fluid increases. In a turbulent flow, the velocity varies dramatically from point to point.

Once the stirring stops, the only force acting on the flow is the viscous shearing force resulting from the velocity gradients present within the fluid. Consequently, the kinetic energy of the flow is *dissipated* by frictional shearing forces such that the velocity decreases and the fluid gradually returns to a steady state.

(Choice A) The incompressibility (or compressibility) of a fluid does not affect the transition from turbulent to laminar flow. Both types of fluids are viscous and can become turbulent.

(Choice B) In many fluid flows, the motion of fluid particles is essentially horizontal such that gravitational potential energy does not play a role in the transition from turbulent to static flow.

(Choice C) **Convection** refers to the transfer of heat due to the large-scale motion of molecules. There is no transfer of heat by convection as the turbulent flow settles down to a resting state.

Educational objective:

Viscosity is a measure of internal friction in a fluid. Kinetic energy in fluid flows is dissipated by the viscous shear force acting between different layers of the fluid flow.

Time spent: 0 sec

Subject:
Physics

QID: 402660

System:
Fluids

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The blood leaving the aorta reaches the circle of Willis, a collection of arteries that supplies blood to the brain. Assume that it takes 600 ms for blood to reach the circle of Willis and that its average velocity is

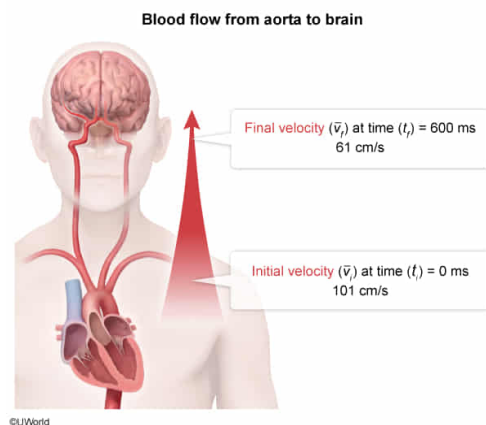
reduced to 61 cm/s. What is the average acceleration experienced by the blood?

- ✓ ☒ A. -67.0 cm/s^2 [46%]
☐ B. -0.670 cm/s^2 [23%]
☐ C. 0.670 cm/s^2 [15%]
☐ D. 67.0 cm/s^2 [14%]

Correct

47% answered correctly

Explanation:



The **acceleration** of an object measures how its velocity v is changing at an instant in time. Because acceleration is derived from velocity, acceleration is also a vector quantity. Evaluating acceleration along one dimension of the xyz-coordinate system over an interval of time (rather than instantaneously) gives the **average acceleration $\bar{a}$** , which is defined as the **change in velocity Δv** over an **interval of time Δt** :

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$$

In this expression, Δv represents the difference between final velocity v_f and initial velocity v_i , and Δt represents the difference between final time t_f and initial time t_i .

If v_f is greater than v_i ($v_f > v_i$), then the average acceleration is **positive** and the **object is accelerating**. Conversely, average acceleration is **negative** when v_i is greater than v_f ($v_i > v_f$).

When blood flows from the aorta to the brain, it ascends the neck structures with negligible horizontal movement; therefore, the blood fluid flow and any associated acceleration occurs essentially along one dimension. Specifically, the v_i of blood is the average velocity of blood within the aorta featured in Table 1:

$$v_i = 101 \text{ cm/s}$$

The blood flow between aorta and brain takes place over 600 ms (0.6 s) and the v_f of blood is 61.0 cm/s. Consequently, the average acceleration of the blood is calculated as:

$$\begin{aligned}\bar{a} &= \frac{61 \frac{\text{cm}}{\text{s}} - 101 \frac{\text{cm}}{\text{s}}}{600 \text{ ms}} \\ \bar{a} &= \frac{-40 \frac{\text{cm}}{\text{s}}}{600 \text{ ms} \cdot \frac{1 \text{ s}}{1000 \text{ ms}}} \\ \bar{a} &= -67.0 \text{ cm/s}^2\end{aligned}$$

Therefore, the blood experiences an acceleration of -67.0 cm/s^2 over 600 ms as they travel from the aorta to the brain (**Choice A**).

Educational objective:

Average acceleration is the change of velocity over a period of time. The average acceleration can be positive or negative, depending on whether the final velocity is larger or smaller than the initial velocity.

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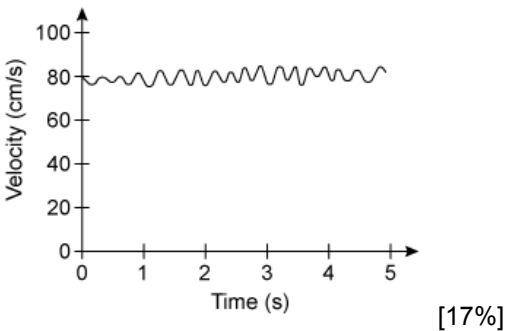
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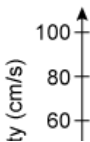
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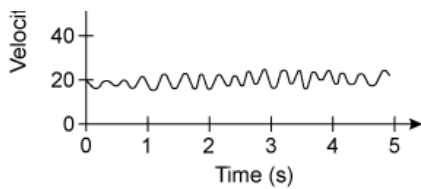
Which graph depicts the flow with the largest turbulent kinetic energy?

☐ A.



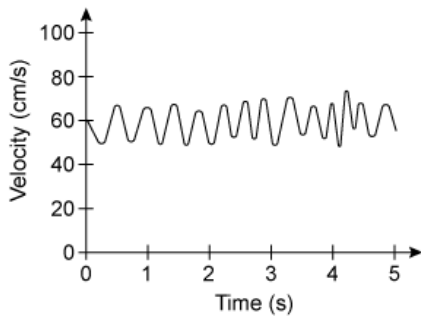
☐ B.





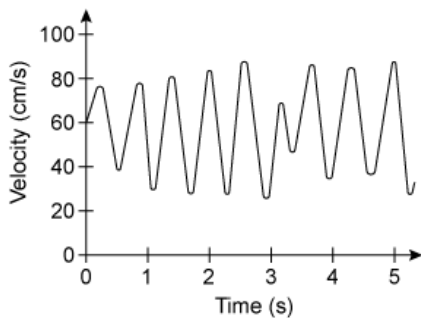
[1%]

☐ C.



[2%]

✓ ☒ D.

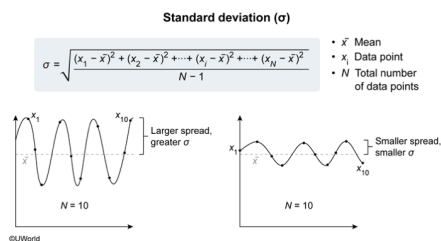


[79%]

Correct

78% answered correctly

Explanation:



Mean and standard deviation are two important ways to characterize a set of data. The **mean (or average)** $\bar{x}$ indicates the central value of the data set, and the **standard deviation** σ describes **how dispersed** the data points are around this central value.

For a sample with some number N of data points x_i , where the subscript index i labels each data point from 1 to N , the value of σ is determined by summing the square of the deviation $(x_i - \bar{x})$ of each data point from the mean, dividing this sum by $(N - 1)$, and then taking the square root of this value:

$$\sigma = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_N - \bar{x})^2}{N - 1}}$$

Two sets of data may have a different σ despite having the same mean because the standard deviation is influenced by the relative spread of values from the mean and not the value of the mean itself.

Turbulent kinetic energy (TKE) is defined in the passage to be proportional to the square of the standard deviation of velocity σ_v :

$$TKE = \frac{1}{2} \sigma_v^2$$

Therefore, the flow with the largest turbulent kinetic energy has the largest average spread in the fluctuations about its mean value. This behavior is best represented by the graph in **Choice D**.

(Choice A) This graph depicts a flow with the highest mean velocity. However, TKE is small because it is proportional only to the average magnitude of the fluctuations about the mean.

(Choice B) The flow depicted in this graph has both a low mean velocity and a low TKE because the position of the mean is low on the y -axis and the fluctuations about the mean are small.

(Choice C) This graph does not depict maximal TKE because the fluctuations about the mean are smaller than those featured in the graph in Choice D.

Educational objective:

Standard deviation is a measure of how much the points in a data set tend to deviate from the mean value. Standard deviation is indicated by the average magnitude of the fluctuations about the mean in a graph.

Time spent:0 sec
Subject:
Physics

QID:402662
System:
Fluids